

Bridging Advanced AI Agent Automation with Enterprise HIPAA Compliance

Seamless Clinecove DB Integration

Clinecove orchestrates the secure execution of state-of-the-art AI agents directly against live clinical trial databases. By integrating the resilient StrangeLoop execution engine with a deterministic HTTP tool boundary, Clinecove automates repetitive operational tasks—such as patient scheduling, study configuration, and data-plane synchronization—without exposing sensitive infrastructure to external networks.

Uncompromising PHI Scope Isolation

At the core of Clinecove's architecture is a multi-layered, secure memory perimeter. Using deterministic data-plane scope predicates, the platform enforces strict tenant-level isolation on all semantic, episodic, and session memories. Records under specific research boundaries (e.g., Study SL-TEST-A, Patient P1) are completely isolated from sibling records or foreign studies (e.g., SL-TEST-B), strictly executing the HIPAA-mandated deny-by-default principle.

Durable & Self-Correcting Execution

Built on a fault-tolerant Elixir/OTP state machine, Clinecove degrades honestly rather than hallucinating when tools fail. External API timeouts and errors are handled transparently. If a Large Language Model produces a malformed payload that violates the target `inputSchema`, the engine's built-in self-correction loop catches the schema mismatch, rejects the malformed input, and triggers immediate model self-correction within milliseconds to maintain clinical data integrity.



Durable OTP Resilience

An active RunReaper watchdog prevents zombie tasks. In the event of an unexpected server restart, all orphaned runs are cleanly swept to a terminal failed state to ensure zero orphaned background threads.

SSRF & Network Vaults

Engine-level SSRF guards block malicious loopback requests by default. Robust authorization tokens and empty scope rejections prevent unauthorized data egress and protect patient privacy.

Real-Time Traceability

Every agent step is audited. Monotonic event traces and live Server-Sent Events (SSE) offer clinical operations teams a complete, real-time window into decision pathways and execution runs.